XIAO JIN

Senior Research Fellow · SIA–NUS Digital Aviation Corporate Laboratory, National University of Singapore

xiao.j@nus.edu.sg

Google Scholar: scholar.google.com/citations?user=NffLyjoAAAAJ · Web: farkasle.github.io

RESEARCH INTERESTS

Simulation optimization and ranking & selection under heteroscedastic noise; contextual and prescriptive analytics; revenue management, dynamic pricing and ancillary pricing; discrete-choice modelling with deep learning; stochastic service operations (aviation, healthcare, maritime, logistics). My work connects methodological contributions in simulation-based decision making to systems that run in production with commercial and clinical partners.

EDUCATION

National University of Singapore — Ph.D., Industrial Systems Engineering & Management *2015.8 – 2020.5*

- Dissertation: *On Solving the Real-Time Decision Problem by Simulation Analytics and Parallelization Acceleration*. Advisor: Prof. Loo Hay Lee. GPA 4.21 / 5.00.

Nanjing University — B.A., School of Management and Engineering *2011.9 – 2015.6*

- GPA 85 / 100.

ACADEMIC APPOINTMENTS

Senior Research Fellow, SIA–NUS Digital Aviation Corporate Laboratory, NUS *2024.11 – present*

Research Fellow, SIA–NUS Digital Aviation Corporate Laboratory, NUS *2022.11 – 2024.11*

Research Fellow, Centre for Next Generation Logistics (C4NGL / C4NGP), NUS *2020.10 – 2022.11*

Research Engineer, C4NGP, NUS *2019.8 – 2020.10*

PUBLICATIONS

Refereed journal articles

[1] Jin, X., Shen, Y., & Lee, L. H. "Contextual Optimizer through Neighborhood Estimation for Prescriptive Analytics." *INFORMS Journal on Computing*, 2026 (Articles in Advance). https://doi.org/10.1287/ijoc.2025.1134

Under review

[2] Jin, X., Ren, J., Sun, H., Liu, C., & Teo, C.-P. "Pricing Checked Baggage under Rational Inattention: Evidence, Estimation, and Field Deployment." *Under review, Operations Research.*

- Randomized price experiments over 1.41M booking sessions across five baggage tiers; a rational-inattention demand model yields an explicit price ladder that scales to thousands of segments. In randomized field deployment the policy raises baggage revenue ~8% ($p < 0.01$) while the posted-price distribution first-order stochastically dominates the incumbent — prices fall on average as revenue rises. Deployed in production at a partner airline.

Working papers

[3] Liang, K., Lu, Y., Mao, J., Sun, S., Yang, C., Zeng, C., Jin, X., Qin, H., Zhu, R., & Teo, C.-P. "Large-Scale Optimization Model Auto-Formulation: Harnessing LLM Flexibility via Structured Workflow." arXiv:2601.09635, 2026.

[4] Liu, C., Sun, H., Teo, C.-P., & Jin, X. "Pricing Analytics for Ancillary Products: Learning from Conversion Data to Optimize Revenue." SSRN 5285522, 2025.

[5] Jin, X., Shu, S., & Teo, C.-P. "Staffing Ahead of Congestion: Certified Reliability for Queues That Do Not Reset." *In preparation.*

- Relocate–Certify–Repair staffing for non-resetting queues with period-level reliability constraints, with an exact finite-sample certificate. On hospital call-centre data, certified plans meet the abandonment target in 565 of 570 weekday-period cells; a field pilot confirmed practical use.

Refereed conference proceedings

[6] Jin, X., Li, H., Lee, L. H., & Chew, E. P. "Optimal Computing Budget Allocation via Sampling-Based Unbiased Approximation." *Proceedings of the Winter Simulation Conference (WSC)*, 2017.

[7] Jin, X. "Budget Allocation Problem in Simulation Analytics." *Proceedings of the Winter Simulation Conference (WSC)*, 2018.

[8] Jin, X., Li, H., & Lee, L. H. "Optimal Budget Allocation in Simulation Analytics." *IEEE International Conference on Automation Science and Engineering (CASE)*, 2019.

[9] Jin, X., Shen, Y., Lee, L. H., Chew, E. P., & Shoemaker, C. A. "A Hybrid of Shrinking Ball Method and Optimal Large Deviation Rate Estimation in Continuous Contextual Simulation Optimization with Single Observation." *WSC*, 2020.

[10] Jin, X., Shen, Y., Li, H., Lee, L. H., Chew, E. P., & Shoemaker, C. A. "Combining Adaptive Budget Allocation with Surrogate Methodology in Solving Continuous Scenario-Based Simulation Optimization." *IEEE CASE*, 2020.

[11] Li, H., Cao, X., Jin, X., Lee, L. H., & Chew, E. P. "Three Carriages Driving the Development of Intelligent Digital Twins — Simulation Plus Optimization and Learning." *WSC*, 2021.

[12] Liu, H., Jin, X., Li, H., Lee, L. H., & Chew, E. P. "A Unified Offline-Online Learning Paradigm via Simulation for Scenario-Dependent Selection." *WSC*, 2021.

[13] Wang, G., Jin, X., & Lee, L. H. "An Efficient Bi-Fidelity Method for Continuous Simulation Optimization." *IEEE CASE*, 2022.

[14] Zhao, T., Jin, X., & Lee, L. H. "Optimal Computing Budget Allocation for Multi-Objective Ranking and Selection Under Bernoulli Distribution." *WSC*, 2022.

[15] Li, D., Liu, H., Jin, X., Li, H., Chew, E. P., Tan, K. C., & Lin, Y. H. "Multi-Fidelity Discrete Optimization via Simulation." *WSC*, 2022.

INVITED TALKS AND CONFERENCE PRESENTATIONS

- "Robust Ancillary Pricing for Airlines: When a Single Mark-up Outperforms Personalisation." **INFORMS Annual Meeting**, Atlanta, GA, 2025.
- Invited speaker, **Sabre** — simulation methodology for revenue-management practitioners, 2025.
- **IORA Industrial Day**, NUS — ancillary pricing research, 2025.
- Presenter, **Winter Simulation Conference** (2017, 2018, 2020) and **IEEE CASE** (2019, 2020).

RESEARCH LEADERSHIP AND FUNDED PROJECTS

- **Lead researcher, SIA–NUS Corp Lab Phase 1** (Revenue Management & Network Planning) — set and drove the research agenda and delivery across the RM and network-planning workstreams. Named researcher on the **SIA–NUS Digital Aviation Corporate Laboratory** award, supported by the **National Research Foundation (NRF), Singapore**.
- **Checked-baggage ancillary pricing, partner airline** (project lead) — rational-inattention pricing policy validated by randomized field deployment (~8% baggage revenue lift, $p < 0.01$) and now running in production.
- **Airline revenue-prediction platform** (lead researcher) — simulation platform over 1-week to 1-year selling horizons with competitor pricing, RM policy and network-level capacity competition; a Transformer-based deep choice model trained on browsing and transaction data drives purchase prediction. Adopted by the airline's practitioners for pre-deployment policy validation.
- **Call-centre rostering, NUHS JMC** (project lead) — simulation–optimization rostering model; call-abandonment rate reduced from 6% to 4%; in live use for monthly staff planning.
- **Appointment scheduling, National Heart Centre Singapore** (project lead) — scheduling under demand uncertainty to uphold the centre's lead-day service promise.
- **Tuas Port Finger 1 capacity study** (project lead) — full-scale discrete-event simulation of ship arrivals, container profiles, crane and AGV dynamics to estimate annual TEU throughput under candidate layouts; findings endorsed by the project stakeholder.
- **SKU repositioning, Kuehne+Nagel** (core researcher) — simulation-based optimization with data-driven demand forecasting for inventory repositioning across warehouses; ~18–21% projected cost reduction.

TEACHING AND MENTORING

Teaching

- Guest lecturer and seminar speaker on simulation optimization and revenue management, NUS.
- Teaching Assistant, Department of Industrial Systems Engineering & Management, NUS (during Ph.D., 2015–2020) — tutorials and grading.

Research mentoring

- Onboarded and technically trained a new research team member into the SIA–NUS Corp Lab revenue-management group — structured ramp-up on the lab's pricing and simulation methods, codebase and data pipeline.

- Directed the work of research interns and assistants on project deliverables.
- Day-to-day technical guide for the lab on deep-learning choice modelling, discrete-event simulation and pricing optimization.

PROFESSIONAL SERVICE

- Referee, *Applied Operations and Analytics* (Taylor & Francis).
- Coordinated the SIA–NUS Corp Lab Phase 1 portfolio — milestone tracking, deliverables and stakeholder reporting across workstreams.
- Industry engagement with the lab's airline partners to communicate research outcomes and scope new use-cases.

TECHNICAL SKILLS

Programming — Python (primary), C, C#, MATLAB, LaTeX

Machine learning — PyTorch, deep learning and Transformers, deep discrete-choice modelling, scikit-learn, XGBoost, pandas, NumPy

Optimization — Gurobi, CPLEX; stochastic and simulation-based optimization, ranking & selection

Simulation — discrete-event simulation (custom C# / C++ engines), SimPy

Data & infrastructure — SQL, Git, Docker, AWS / GCP; Django, React; AI-assisted development (Claude Code)

REFERENCES

Available on request.